

Quiz details

Quiz instructions:

This quiz will focus on the material from Week 8 in the lab workbook. To answer the following questions, you will need to use the [power_up](https://canvas.stir.ac.uk/courses/19303/files/4997332/download?download_frd=1) ↓ (https://canvas.stir.ac.uk/courses/19303/files/4997332/download?download_frd=1) dataset from [Week 4](#). This practical included categorical data (e.g., mobile app OS and dam type), so we can use it to practice ANOVAs and associated tests.

☒ Show question details

Question 1 1 pts

Assume that you want to run a ANOVA to test the null hypothesis that mean player score is not different for different dam sizes. Before running the ANOVA, use a Levene's test to test the null hypothesis that group variances are the same. Based on the Levene's test, what should you conclude?

- ☐ Do not reject the null hypothesis of equal variances because $F > 0.05$
- ☐ Reject the null hypothesis of equal variances because $F < 0.05$
- ☐ Do not reject the null hypothesis because $P > 0.05$
- ☐ Reject the null hypothesis because $P < 0.05$
- ☐ None of the above

Correct answer comments

A Levenes homogeneity of variance test can be used by going to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Score' into the Dependent Variables box and 'Dam_size' into the Grouping variables box. Next, check the 'Homogeneity test' box under 'Assumption Checks'. The Homogeneity of variances test will appear in the right panel of Jamovi. The p-value of the test is 0.514, so we do not reject the null hypothesis that variances are equal.

Wrong answer comments

A Levenes homogeneity of variance test can be used by going to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Score' into the Dependent Variables box and 'Dam_size' into the Grouping variables box. Next, check the 'Homogeneity test' box under 'Assumption Checks'. The Homogeneity of variances test will appear in the right panel of Jamovi. The p-value of the test is 0.514, so we do not reject the null hypothesis that variances are equal.

Question 2 1 pts

Run the most appropriate test to test the null hypothesis that mean player score is not different for different dam sizes. What is the test statistic calculated from the test? Report your answer to 2 or more decimal places.

Between 0.77 and 0.78

Correct answer comments

The Levene's test from question 1 confirms that we can run a one-way ANOVA instead of the non-parametric equivalent. Go to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Score' into the Dependent Variables box and 'Dam_size' into the Grouping variables box. Next, make sure that 'Assume equal (Fisher's)' is checked under **Variances**. An ANOVA output table will appear to the right with an test statistic of $F = 0.77680$.

Wrong answer comments

The Levene's test from question 1 confirms that we can run a one-way ANOVA instead of the non-parametric equivalent. Go to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Score' into the Dependent Variables box and 'Dam_size' into the Grouping variables box. Next, make sure that 'Assume equal (Fisher's)' is checked under **Variances**. An ANOVA output table will appear to the right with an test statistic of $F = 0.77680$.

Question 3 1 pts

Based on the output of the test in Question 2, what conclusion should we make?

- ☐ Reject the null hypothesis that dam sizes have the same mean score
- ☐ Do not reject the null hypothesis that dam sizes have the same mean score
- ☐ Accept the null hypothesis that dam sizes have the same mean score
- ☐ Reject the null hypothesis that dam sizes have different mean scores
- ☐ None of the above

Correct answer comments

The output table of the one-way ANOVA has a p-value of $P = 0.464$. Because this value is greater than 0.05, we do not reject the null hypothesis that group means are equal (i.e., that dam sizes have the same score). Note that we should never accept a null hypothesis (see [Chapter 20.3](https://bradduthie.github.io/SCIU4T4/Chapter_20.html#p-values-false-positives-and-power) [↗\(https://bradduthie.github.io/SCIU4T4/Chapter_20.html#p-values-false-positives-and-power\)](https://bradduthie.github.io/SCIU4T4/Chapter_20.html#p-values-false-positives-and-power)).

Wrong answer comments

The output table of the one-way ANOVA has a p-value of $P = 0.464$. Because this value is greater than 0.05, we do not reject the null hypothesis that group means are equal (i.e., that dam sizes have the same score). Note that we should never accept a null hypothesis (see [Chapter 20.3](https://bradduthie.github.io/SCIU4T4/Chapter_20.html#p-values-false-positives-and-power) [↗\(https://bradduthie.github.io/SCIU4T4/Chapter_20.html#p-values-false-positives-and-power\)](https://bradduthie.github.io/SCIU4T4/Chapter_20.html#p-values-false-positives-and-power)).

Question 4 1 pts

Run a two-way ANOVA to test whether or not OS, Dam_size, and their interaction have a significant effect on player score. Based on the ANOVA output table, which of the following null hypotheses should be rejected?

- ☐ Mean player score does not differ by OS
- ☐ Mean player score does not differ by Dam size
- ☐ There is no interaction between OS and Dam size

- ☐ We should reject more than one null hypothesis
- ☐ None of the above

Correct answer comments

From the Analyses tab in Jamovi, navigate to ANOVA, then choose ANOVA again in the pulldown menu. Put OS and Dam_size in the Fixed Factors box, then Score in the Dependent Variable box. From the ANOVA output table, the only p-value less than 0.05 is from the test of the effect of OS on player score ($P = 0.022$). The remaining p-values are greater than 0.05, so the only null hypothesis that we should reject is that player score is not different for different OS types.

Wrong answer comments

From the Analyses tab in Jamovi, navigate to ANOVA, then choose ANOVA again in the pulldown menu. Put OS and Dam_size in the Fixed Factors box, then Score in the Dependent Variable box. From the ANOVA output table, the only p-value less than 0.05 is from the test of the effect of OS on player score ($P = 0.022$). The remaining p-values are greater than 0.05, so the only null hypothesis that we should reject is that player score is not different for different OS types.

Question 5 1 pts

Run the same two-way ANOVA from Question 4 to test whether or not OS, Dam_size, and their interaction have a significant effect on player score. What is the sum of squares for the unexplained variation in the ANOVA? Report your answer to 1 or more decimal places.

32,666.6428 (with margin: 2)

Correct answer comments

From the Analyses tab in Jamovi, navigate to ANOVA, then choose ANOVA again in the pulldown menu. Put OS and Dam_size in the Fixed Factors box, then Score in the Dependent Variable box. The sum of squares for the unexplained variation is just the residual sum of squares from the ANOVA output table. This has a value of 32666.64278.

Wrong answer comments

From the Analyses tab in Jamovi, navigate to ANOVA, then choose ANOVA again in the pulldown menu. Put OS and Dam_size in the Fixed Factors box, then Score in the Dependent Variable box. The sum of squares for the unexplained variation is just the residual sum of squares from the ANOVA output table. This has a value of 32666.64278.

Question 6 1 pts

Now suppose that you want to test the null hypothesis that mean Biodiversity investment does not differ with Dam size. Test the assumptions of ANOVA. Based on these assumptions, what kind of test should you use to test the null hypothesis?

- ☐ One-way ANOVA
- ☐ Two-way ANOVA
- ☐ ANCOVA
- ☐ Wilcoxon Signed Rank test
- ☐ Kruskal-Wallis Test

Correct answer comments

A Shapiro-Wilk test of normality and Levene's homogeneity of variance test can be used by going to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Biodiversity' into the Dependent Variables box and 'Dam_size' into the Grouping variables box. Next, check the 'Homogeneity test' and 'Normality test' boxes under 'Assumption Checks'.

If we check the assumptions of ANOVA, we can see that a test of normality (Shapiro-Wilk) shows that we should reject the null hypothesis that Biodiversity investment is normally distributed ($P < 0.001$). Biodiversity investment is also an integer, which should make us suspicious of the assumption of normality. While ANOVA is robust to deviations of normality, here we should use the non-parametric Kruskal-Wallis test instead.

Wrong answer comments

A Shapiro-Wilk test of normality and Levene's homogeneity of variance test can be used by going to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Biodiversity' into the Dependent Variables box and 'Dam_size' into the Grouping variables box. Next, check the 'Homogeneity test' and 'Normality test' boxes under 'Assumption Checks'.

If we check the assumptions of ANOVA, we can see that a test of normality (Shapiro-Wilk) shows that we should reject the null hypothesis that Biodiversity investment is normally distributed ($P < 0.001$). Biodiversity investment is also an integer, which should make us suspicious of the assumption of normality. While ANOVA is robust to deviations of normality, here we should use the non-parametric Kruskal-Wallis test instead.

Question 7 1 pts

Run the appropriate test of the null hypothesis that mean Biodiversity investment does not differ with Dam size. Based on the output, what is the most accurate conclusion?

- ☐ The probability that the null hypothesis is true is $P = 0.021$.
- ☐ y that the null hypothesis is false is $P = 0.021$. The probabilit
- ☐ If the null hypothesis is true, then the probability of getting our Chi-square value or higher is $P = 0.021$.
- ☐ If the null hypothesis is true, then the probability of getting our Chi-square value or lower is $P = 0.021$.
- ☐ The probability that the null hypothesis is true is $P = 0.082$.
- ☐ The probability that the null hypothesis is false is $P = 0.082$.
- ☐ If the null hypothesis is true, then the probability of getting as high or higher than our F is 0.082.
- ☐ If the null hypothesis is true, then the probability of getting as low or lower than our F is 0.082.

Correct answer comments

The correct test to run here is the Kruskal-Wallis test. Go to the Analysis tab of the Jamovi toolbar and select ANOVA, then 'One-Way ANOVA: Kruskal-Wallis' from the pulldown menu (second to last option). Place 'Biodiversity' into the Dependent Variables box and 'Dam_size' into the Grouping variables box.

Remember that a p-value is the probability of getting a test statistic as or more extreme than the one we calculated, assuming that the null hypothesis is true. We cannot know the probability that the null hypothesis (or the alternative) is true (see [Chapter 20.1 \(https://bradduthie.github.io/SCIU4T4/Chapter_20.html#how-ridiculous-is-our-hypothesis\)](https://bradduthie.github.io/SCIU4T4/Chapter_20.html#how-ridiculous-is-our-hypothesis)). The p-value of the Kruskal-Wallis test (from the output table under p) is 0.021.

Wrong answer comments

The correct test to run here is the Kruskal-Wallis test. Go to the Analysis tab of the Jamovi toolbar and select ANOVA, then 'One-Way ANOVA: Kruskal-Wallis' from the pulldown menu (second to last option). Place 'Biodiversity' into the Dependent Variables box and 'Dam_size' into the Grouping variables box.

Remember that a p-value is the probability of getting a test statistic as or more extreme than the one we calculated, assuming that the null hypothesis is true. We cannot know the probability that the null hypothesis (or the alternative) is true (see [Chapter 20.1 \(https://bradduthie.github.io/SCIU4T4/Chapter_20.html#how-ridiculous-is-our-hypothesis\)](https://bradduthie.github.io/SCIU4T4/Chapter_20.html#how-ridiculous-is-our-hypothesis)). The p-value of the Kruskal-Wallis test (from the output table under p) is 0.021.

New data

Now assume that there is some new data from a specific population of interest (e.g., undergraduate science students). In total, there are 18 new players sampled ($N = 18$). Six of these players chose a small dam, 6 chose a medium dam, and 6 chose a large dam. Their scores are shown in the table below.

Table of scores for new players of Power Up!

Small	Medium	Large
113.1	101.7	127.0
96.4	86.3	125.1
95.7	87.3	103.8
110.2	69.2	100.4
86.8	77.9	110.9
123.0	107.1	128.6

Use the table above to answer the remaining questions 8-10. To get started, make sure to input these into Jamovi in a tidy format.

Question 8 1 pts

Suppose that you want to test the null hypothesis that players using a small, medium, and large dam all have the same score. Based on your assumption checks of the data, which type of test should you use?

☐ One-way ANOVA

- ☐ Two-way ANOVA
- ☐ ANCOVA
- ☐ Wilcoxon Signed Rank test
- ☐ Kruskal-Wallis Test

Correct answer comments

A Shapiro-Wilk test of normality and Levene's homogeneity of variance test can be used by going to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Score' into the Dependent Variables box and 'Dam_size' into the Grouping variables box. Next, check the 'Homogeneity test' and 'Normality test' boxes under 'Assumption Checks'.

If we check the assumptions of ANOVA, we can see that our Shapiro-Wilk test of normality gives us a p-value of $P = 0.233$, and our Levene's test gives us a p-value of $P = 0.990$. Since neither of these p-values exceed 0.05, we do not reject the null hypothesis that the data are normally distributed, or that the variances are equal. It is therefore safe to use the one-way ANOVA.

Wrong answer comments

A Shapiro-Wilk test of normality and Levene's homogeneity of variance test can be used by going to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Score' into the Dependent Variables box and 'Dam_size' into the Grouping variables box. Next, check the 'Homogeneity test' and 'Normality test' boxes under 'Assumption Checks'.

If we check the assumptions of ANOVA, we can see that our Shapiro-Wilk test of normality gives us a p-value of $P = 0.233$, and our Levene's test gives us a p-value of $P = 0.990$. Since neither of these p-values exceed 0.05, we do not reject the null hypothesis that the data are normally distributed, or that the variances are equal. It is therefore safe to use the one-way ANOVA.

Question 9 1 pts

Run the most appropriate test identified in Question 8 to test the null hypothesis that players using a small, medium, and large dam all have the same score. What is the p-value of this test? Report your answer to 3 decimals.

0.009 (with margin: 0.001)

Correct answer comments

We can run a one-way ANOVA to test the null hypothesis. Go to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Score' into the Dependent Variables box and 'Dam_size' into the Grouping variables box. Check the box for 'Assume equal (Fisher's)' under the **Variances** options. From the one-way ANOVA output, we can see that $P = 0.009$.

Wrong answer comments

We can run a one-way ANOVA to test the null hypothesis. Go to the Analysis tab of the Jamovi toolbar and selecting ANOVA, then 'One-Way ANOVA' from the pulldown menu. Place 'Score' into

the Dependent Variables box and 'Dam_size' into the Grouping variables box. Check the box for 'Assume equal (Fisher's)' under the **Variances** options. From the one-way ANOVA output, we can see that $P = 0.009$.

Question 10 1 pts

Run a Tukey's Honestly Significant Differences (HSD) test to determine which dam types produce significantly different player scores. From this Tukey's HSD, which of the following two dam sizes produce significantly different scores?

- ☐ small and medium
- ☐ small and large
- ☐ medium and large
- ☐ all dam sizes are significantly different
- ☐ no dam sizes are significantly different

Correct answer comments

There are two ways to conduct the Tukey's HSD. One is to go underneath the One-Way ANOVA input to the pulldown called 'Post-Hoc Tests'. You can then check the radio button 'Tukey (equal variances)'. A table will appear in the panel to the right with comparisons between all combinations of small, medium, and large dams. P-values are 0.613 (Large vs Medium), 0.548 (Large vs Small), and 0.998 (Medium vs Small). Since no p-value is less than 0.05, we conclude that no dams are significantly different in score.

Another approach is to set up an ANOVA by going to the ANOVA button in the jamovi toolbar and choosing 'ANOVA' from the pulldown menu. Put dam size as the fixed factor and Score as the dependent variable, then scroll down to the 'Post Hoc Tests' pulldown menu. Select 'Tukey' from the available checkboxes and put dam size in the box to the right. A table of comparisons will appear with a value of p_{tukey} as the last column, none of the p-values are less than 0.05.

Wrong answer comments

There are two ways to conduct the Tukey's HSD. One is to go underneath the One-Way ANOVA input to the pulldown called 'Post-Hoc Tests'. You can then check the radio button 'Tukey (equal variances)'. A table will appear in the panel to the right with comparisons between all combinations of small, medium, and large dams. P-values are 0.613 (Large vs Medium), 0.548 (Large vs Small), and 0.998 (Medium vs Small). Since no p-value is less than 0.05, we conclude that no dams are significantly different in score.

Another approach is to set up an ANOVA by going to the ANOVA button in the jamovi toolbar and choosing 'ANOVA' from the pulldown menu. Put dam size as the fixed factor and Score as the dependent variable, then scroll down to the 'Post Hoc Tests' pulldown menu. Select 'Tukey' from the available checkboxes and put dam size in the box to the right. A table of comparisons will appear with a value of p_{tukey} as the last column, none of the p-values are less than 0.05.